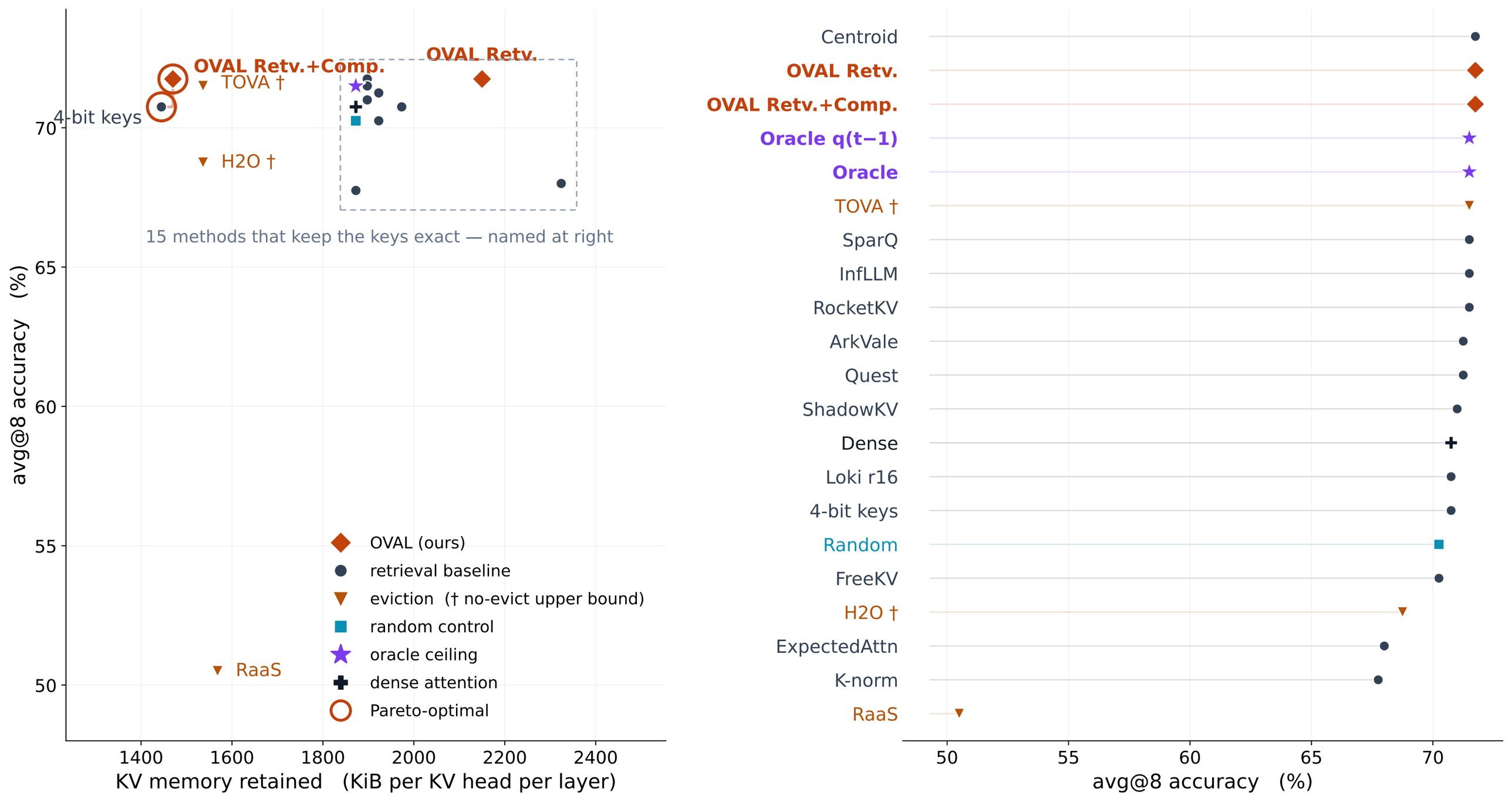
OVAL Retv.+Comp.
OVAL Retv.
TOVA †
4-bit keys
H2O †
15 methods that keep the keys exact — named at right
RaaS
OVAL (ours)
retrieval baseline
eviction († no-evict upper bound)
random control
oracle ceiling
dense attention
Pareto-optimal
avg@8 accuracy (%)
KV memory retained (KiB per KV head per layer)
70
65
60
55
50
1400
1600
1800
2000
2200
2400
Centroid
OVAL Retv.
OVAL Retv.+Comp.
Oracle q(t−1)
Oracle
TOVA †
SparQ
InfLLM
RocketKV
ArkVale
Quest
ShadowKV
Dense
Loki r16
4-bit keys
Random
FreeKV
H2O †
ExpectedAttn
K-norm
RaaS
50
55
60
65
70
avg@8 accuracy (%)